

Introducción

El presente desafío tiene como finalidad aplicar los conocimientos sobre scripting en Linux para automatizar la creación y configuración de una máquina virtual utilizando la herramienta `VBoxManage` de VirtualBox. Esta automatización permite reducir el tiempo necesario para preparar entornos de prueba o desarrollo, asegurando consistencia y rapidez en la configuración de las máquinas virtuales. El script desarrollado recibe parámetros desde la línea de comandos y configura componentes esenciales como CPU, memoria, disco duro y controladores SATA/IDE.

Desarrollo

```
soflob@LinuxVR: ~/Documents/Challenge_1
soflob@LinuxVR:~/Documents/Challenge_1$ ./vm_auto_setup.sh LinuxVM Linux_64 2 4 128 100 Controlador
Virtual machine 'LinuxVM' is created and registered.
UUID: 675d5078-dbba-40e8-9a9d-0434481a67ab
Settings file: '/home/soflob/VirtualBox VMs/LinuxVM/LinuxVM.vbox'
0%...10%...20%...30%...40%...50%...60%...70%...80%...90%...100%
Medium created. UUID: c70237a3-d36b-4a72-9dfe-23bc9bfe608a
-----
Máquina Virtual 'LinuxVM' creada exitosamente:
Name: LinuxVM
Encryption: disabled
Groups: /
Guest OS: Other Linux (64-bit)
UUID: 675d5078-dbba-40e8-9a9d-0434481a67ab
Config file: /home/soflob/VirtualBox VMs/LinuxVM/LinuxVM.vbox
Snapshot folder: /home/soflob/VirtualBox VMs/LinuxVM/Snapshots
Log folder: /home/soflob/VirtualBox VMs/LinuxVM/Logs
Hardware UUID: 675d5078-dbba-40e8-9a9d-0434481a67ab
Memory size: 4096MB
Page Fusion: disabled
VRAM size: 128MB
CPU exec cap: 100%
HPET: disabled
CPUProfile: host
Chipset: piix3
Firmware: BIOS
Number of CPUs: 2
PAE: enabled
Long Mode: enabled
Triple Fault Reset: disabled
APIC: enabled
X2APIC: enabled
Nested VT-x/AMD-V: disabled
CPUID Portability Level: 0
```

```
soflob@LinuxVR: ~/Documents/Challenge_1
CPUID Portability Level: 0
CPUID overrides: None
Boot menu mode: message and menu
Boot Device 1: Floppy
Boot Device 2: DVD
Boot Device 3: HardDisk
Boot Device 4: Not Assigned
ACPI: enabled
IOAPIC: enabled
BIOS APIC mode: APIC
Time offset: 0ms
BIOS NVRAM File: /home/soflob/VirtualBox VMs/LinuxVM/LinuxVM.nvram
RTC: local time
Hardware Virtualization: enabled
Nested Paging: enabled
Large Pages: enabled
VT-x VPID: enabled
VT-x Unrestricted Exec.: enabled
AMD-V Virt. Vmsave/Vmload: enabled
IOMMU: None
Paravirt. Provider: Default
Effective Paravirt. Prov.: KVM
State: powered off (since 2025-04-14T07:56:37.209000000)
Graphics Controller: VBoxVGA
Monitor count: 1
3D Acceleration: disabled
2D Video Acceleration: disabled
Teleporter Enabled: disabled
Teleporter Port: 0
Teleporter Address:
Teleporter Password:
Tracing Enabled: disabled
Allow Tracing to Access VM: disabled
```

```
soflob@LinuxVR: ~/Documents/Challenge_1
Allow Tracing to Access VM: disabled
Tracing Configuration:
Autostart Enabled: disabled
Autostart Delay: 0
Default Frontend:
VM process priority: default
Storage Controllers:
#0: 'Controlador-SATA', Type: IntelAhci, Instance: 0, Ports: 30 (max 30), Bootable
    Port 0, Unit 0: UUID: c70237a3-d36b-4a72-9dfe-23bc9bfe608a
    Location: "/home/soflob/VirtualBox VMs/LinuxVM/LinuxVM.vdi"
#1: 'Controlador-IDE', Type: PIIX4, Instance: 0, Ports: 2 (max 2), Bootable
    Port 0, Unit 0: Empty
NIC 1: MAC: 0800273290DA, Attachment: NAT, Cable connected: on, Trace: off (file: none), Type: 825
40EM, Reported speed: 0 Mbps, Boot priority: 0, Promisc Policy: deny, Bandwidth group: none
NIC 1 Settings: MTU: 0, Socket (send: 64, receive: 64), TCP Window (send:64, receive: 64)
NIC 2: disabled
NIC 3: disabled
NIC 4: disabled
NIC 5: disabled
NIC 6: disabled
NIC 7: disabled
NIC 8: disabled
Pointing Device: PS/2 Mouse
Keyboard Device: PS/2 Keyboard
UART 1: disabled
UART 2: disabled
UART 3: disabled
UART 4: disabled
LPT 1: disabled
LPT 2: disabled
Audio: enabled (Driver: Default, Controller: AC97, Codec: STAC9700)
Audio playback: disabled
Audio capture: disabled
```

```
soflob@LinuxVR: ~/Documents/Challenge_1
UART 4: disabled
LPT 1: disabled
LPT 2: disabled
Audio: enabled (Driver: Default, Controller: AC97, Codec: STAC9700)
Audio playback: disabled
Audio capture: disabled
Clipboard Mode: disabled
Drag and drop Mode: disabled
VRDE: disabled
OHCI USB: disabled
EHCI USB: disabled
xHCI USB: disabled
USB Device Filters: <none>
Bandwidth groups: <none>
Shared folders: <none>
Recording enabled: no
Recording screens: 1
Screen 0:
  Enabled: yes
  ID: 0
  Record video: yes
  Destination: File
  File: /home/soflob/VirtualBox VMs/LinuxVM/LinuxVM-screen0.webm
  Options: vc_enabled=true,ac_enabled=false,ac_profile=med
  Video dimensions: 1024x768
  Video rate: 512kbps
  Video FPS: 25fps
* Guest:
Configured memory balloon: 0MB
-----
soflob@LinuxVR:~/Documents/Challenge_1$
```

Conclusión

La automatización mediante Bash y `VBoxManage` proporciona una solución eficiente para la creación de entornos virtuales en VirtualBox. A través de esta actividad se logró comprender la estructura de comandos necesarios para crear y configurar máquinas virtuales desde la terminal, sin intervención manual en la interfaz gráfica. Esta práctica fortalece habilidades esenciales en scripting, administración de sistemas y virtualización, aplicables a entornos reales de TI y DevOps.